Shrikaran P

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Professional Summary

Forward-thinking Engineering student with a strong foundation in **Embedded Systems, Signal Processing, and Deep Learning**. Hands-on experience developing preprocessing pipelines and fine-tuning **CNN models** for real-world data. Proficient in **C, C++, and Python,** with a proven track record of bridging hardware-software gaps through **IoT development and EEG signal analysis**. Eager to contribute to Opto-ML's mission of advancing AI/ML semiconductor technology through low-level firmware optimization and hardware-software co-design.

Technical Skills

Programming Languages: C, C++, Python, Java.
AI/ML & Data: AI/ML Applications, CNN-LSTM Modeling, PyTorch, MySQL.
Tools & Environment: Git/Version Control, Full-Stack Development.
Core Competencies: Problem Solving, Analytical Thinking, Team Collaboration.

Educational Qualifications

B.Tech-Computer science engineering(5th semester) May 2023 - May 2027
SRM Institute of science and Technology - Tiruchchirappalli
CGPA - 9.2/10

Professional Credentials & Experience

EEG Signal Analysis & Motor Imagery Classification Internship – NIT Tiruchirappalli
Foundations of AI/ML – Geeks for Geeks
AWS Academy Cloud Foundations – AWS
Java Full Stack Development – EduSkill

Technical Projects

EEG Signal Analysis & Motor Imagery Classification (NITT Internship)

- **Developed and fine-tuned CNN models** using **PyTorch** for high-accuracy motor imagery classification on real-world datasets.
- Engineered complex preprocessing pipelines incorporating **bandpass filtering, STFT (Short-Time Fourier Transform), and artifact removal** to optimize signal-to-noise ratios.
- **Optimized model performance for healthcare and Brain-Computer Interface (BCI) applications**, leading to a co-authored research paper on practical ML implementations.

Deep Learning-based Intrusion Detection System

- **Architected a hybrid CNN-LSTM model** to detect and classify malicious network traffic using the **CICIDS2017** dataset.
- **Implemented real-time monitoring logic** to enhance threat classification speed and accuracy, focusing on reducing computational latency.

Research & Publications

A Secure On-Premises ETL Pipeline Using OCR and Local LLMs - ICCMC 2025 {Link}

- **Engineered a private ETL pipeline integrating OCR modules** with locally hosted **Large Language Models (Ollama)** to process unstructured enterprise data.
- **Optimized for on-premises deployment,** ensuring strict data privacy and regulatory compliance (GDPR, HIPAA) while significantly increasing extraction efficiency.
- Demonstrates experience with local AI deployment and optimizing software to run efficiently without cloud reliance—directly applicable to **edge-AI hardware**.

Artificial Intelligence in Smart Agriculture | Indian Journal of Natural Sciences (Oct 2024)

- **Developed and deployed machine learning models** for precision agriculture, resulting in a **30% improvement** in yield prediction accuracy.
- **Published research** detailing the practical application of AI in optimizing resource allocation and predictive analytics in complex environments.

Languages: Tamil,English

References: Vishnu Priya, NIT Tiruchirappalli | Email:vishnupriya@nitt.edu | Phone:+91 9486072557